

MAUS User Guide

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Note: This is a placeholder (“dummy”) guide scaffold. Replace the bracketed [TODO] sections with project-specific content before publishing.

Methyl Assignments Using Satisfiability — a clean-room reimplementaion of the SAT-based methyl NMR assignment method of Nerli et al., *Nat. Commun.* 12:691 (2021).

1. Overview

MAUS assigns methyl NMR resonances by **subgraph isomorphism** rather than by scoring. Given a protein structure and a set of 2D reference peaks with NOE connectivity, it returns — for each peak — the *set* of methyls consistent with every hard constraint. It never emits a single guess unless the data force one, and it provably never excludes a valid assignment.

Concept	In this tool
Structure graph G	one node per methyl carbon; geminal / short / long edges
Data graph H	2D peaks (residue type known) + symmetric NOE edges
Solver	python-sat (Glucose3), assumption-based enumeration
Output	per-peak option set (1 / 2-3 / >3 / unassigned)

2. Installation

```
python3 -m pip install python-sat
git clone https://github.com/deepnrmr/maus.git
cd maus
```

Requirements:

- Python ≥ 3.8

- python-sat (provides the Glucose3 SAT solver)
-

3. Quick start

```
python maus.py examples/mbp/1ANF.pdb examples/mbp/mbp_peaks.tsv \
  --keep-k 8 --out mbp_options.tsv
```

Expected summary:

```
methyIs(G nodes)=192  peaks=192
unique(1 option)      = 176/192
ambiguous(2-3 options)= 16/192
unassigned            = 0/192
truth in option set   = 192/192 = 100.0%
```

4. Input formats

4.1 PDB structure

Standard ATOM records. Only these residue types are parsed by default: ALA ILE LEU MET THR VAL. Chain A (or blank) is used.

4.2 Peaks TSV

Tab-separated, one peak per line:

```
peak_id <TAB> residue_type <TAB> truth_label
P1      L              L7CD1
P2      L              L7CD2
```

- peak_id — arbitrary unique string
 - residue_type — one-letter code (A I L M T V)
 - truth_label — ground-truth methyl label (for benchmarking only)
-

5. Command-line options

Option	Default	Meaning
--short-cut	6.0	structure short-range edge cutoff (Å)
--long-cut	10.0	structure long-range edge cutoff (Å)
--noe-short	6.0	simulated short-mixing NOE cutoff (Å)
--noe-long	8.0	simulated long-mixing NOE cutoff (Å)
--keep-k	8	nearest-K NOE partners kept per methyl

Option	Default	Meaning
--labeling	A;I;L;M;T;V	residue types present
--out	–	write per-peak options TSV

6. Interpreting output

The --out TSV columns:

Column	Meaning
peak_id	input peak identifier
res_type	one-letter residue type
n_options	number of valid methylys
options	comma-separated methyl labels
truth	ground-truth label
truth_in_set	1 if truth $\in$ options, else 0

- **1 option** $\rightarrow$ uniquely assigned.
- **2–3 options** $\rightarrow$ residual ambiguity, usually geminal/local symmetry.
- **>3 options** $\rightarrow$ weak local NOE network.
- **0 options** $\rightarrow$ over-constrained inputs (check cutoffs).

7. Caveat on the bundled benchmark

Important. In the shipped MBP example the NOE data graph **H** is *simulated from the same coordinates* used to build **G**, and peaks are indexed identically to methylys. The truth in option set = 100% figure is therefore a self-consistency check on ideal, noise-free, complete data — not a measurement of real-data assignment performance. To reproduce the paper’s validation, supply an **independent** experimental NOE peak list.

8. Troubleshooting

Symptom	Likely cause	Fix
ModuleNotFoundError: pysat	solver not installed	pip install python-sat
many unassigned peaks	cutoffs too tight	raise --long-cut / --noe-long
everything ambiguous	NOE network too sparse	raise --keep-k

9. References

- Nerli, De Paula, McShan & Sgourakis. *Backbone-independent NMR resonance assignments of methyl probes in large proteins*. **Nat. Commun.** 12:691 (2021). [doi:10.1038/s41467-021-20984-0](https://doi.org/10.1038/s41467-021-20984-0)
- See COMPARISON.md for the head-to-head against MAGIC.

[TODO] Add author, license, and contact sections before release.